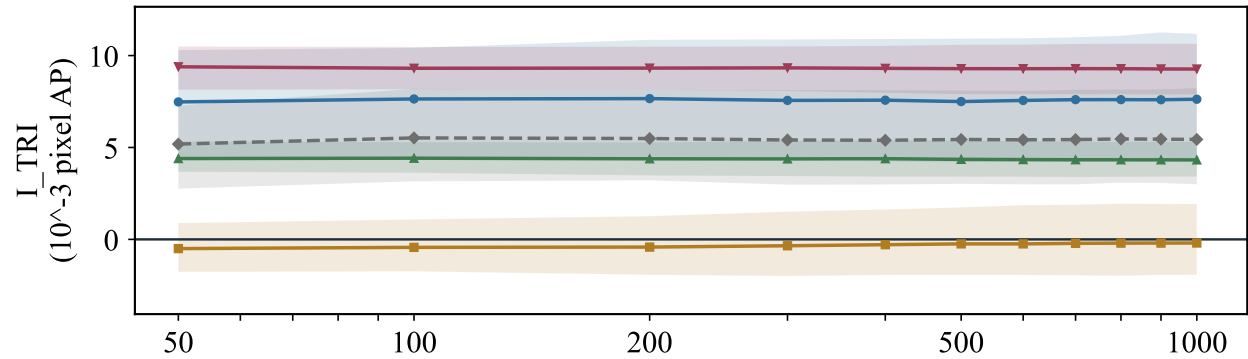


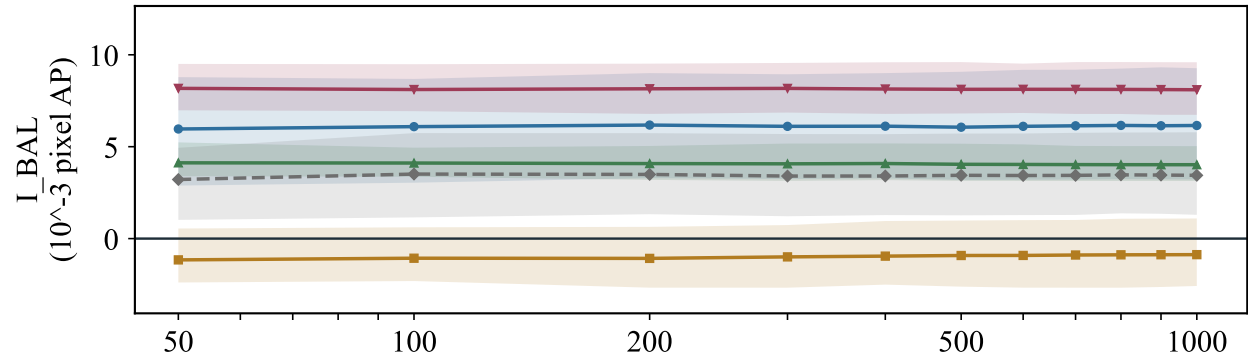
Figure S4. Bootstrap convergence of the interaction: the reported estimates and intervals have settled.

MPDD BTAD MVTec VisA KSDD2

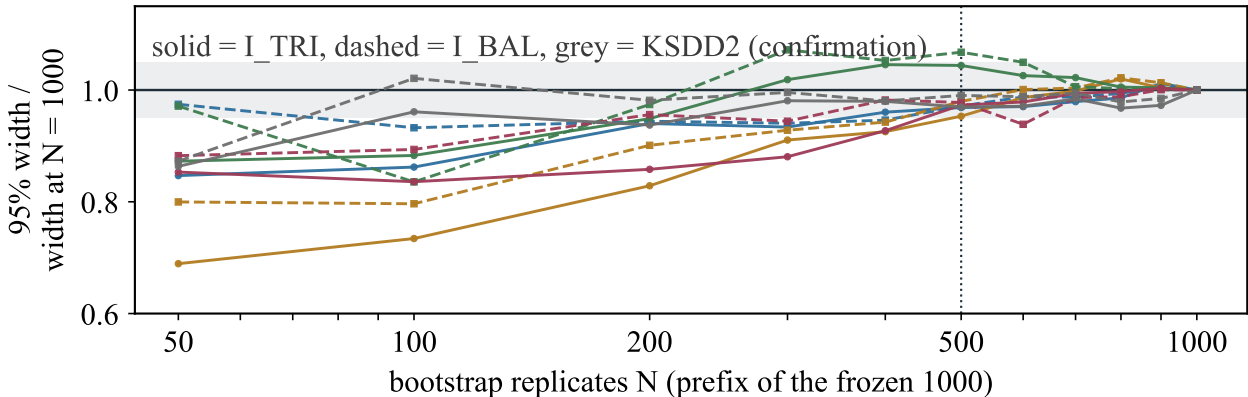
(a) I_TRI: running estimate and 95% interval vs. bootstrap replicates



(b) I_BAL: the same, for a different split of the slot weights



(c) 95% interval width relative to its value at N = 1000



No target-domain training: no optimisation objective, hence no loss-versus-iteration curve.
Prefix bootstrap: the first N of the 1000 stored draws are reused; no new sampling.
Estimates: $N \geq 200$ moves all 10 series by $\leq 2.2 \times 10^{-4}$ pixel AP; $N \geq 500$ by $\leq 1.3 \times 10^{-4}$.
Interval width: $N \geq 500$ stays within 6.8% of its $N = 1000$ value ($N = 200$: 17.1%).
So the frozen 1000 replicates are reported; endpoints converge slower than the estimate.
Roles: MPDD development, BTAD holdout, MVTec/VisA external frozen validation.
KSDD2 is the confirmation set, outside the four-dataset family (F_SPEC.json decision C).
Sources and the full prefix table: figS4_bootstrap_convergence.json.